(i) Distribution function

The distribution function $F(x) = P(X \le x) = \int_{-\infty}^{x} f(u)\,du$.

When $x < 1$, $F(x) = \int_{-\infty}^{x} f(u)\,du = \int_{-\infty}^{x} 0\,du = 0$.

When $1 \le x < 5$ $F(x) = \int_{-\infty}^{x} f(u)\,du = \int_{-\infty}^{1} 0\,du + \int_{1}^{x} \frac{1}{4}\,du = \frac{1}{4}(x-1)$.

When $x \ge 5$ $F(x) = \int_{-\infty}^{x} f(u)\,du = \int_{-\infty}^{1} 0\,du + \int_{1}^{5} \frac{1}{4}\,du + \int_{5}^{x} 0\,du = 1$.

Thus
$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{x-1}{4}, & 1 \le x \le 5 \\ 1, & x > 5. \end{cases}$$

(ii) $P(X < 3) = P(X \le 3) = F(3) = \frac{3-1}{4} = \frac{1}{2}$ (since $F(x)$ is continuous).

(iii) $P(2 < X < 4) = P(2 \le X \le 4) = F(4) - F(2) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$.

(iv) $P(3 \le X) = P(X \ge 3) = 1 - P(X < 3) = 1 - \frac{1}{2} = \frac{1}{2}$.

Example 11.15

Let X be a random variable denoting the life time of an electrical equipment having probability density function

$$f(x) = \begin{cases} k\,e^{-2x} & \text{for } x > 0 \\ 0 & \text{for } x \le 0. \end{cases}$$

Find (i) the value of k (ii) Distribution function (iii) $P(X < 2)$

(iv) calculate the probability that X is at least for four unit of time (v) $P(X = 3)$.

Solution

(i) Since $f(x)$ is a probability density function, $f(x) \ge 0$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$

That is $\int_{-\infty}^{0} 0\,dx + \int_{0}^{\infty} k\,e^{-2x}dx = 1.$

$$0 + k\left(\frac{e^{-2x}}{-2}\right)_0^{\infty} = 1 \Rightarrow k\left(\frac{e^{-\infty} - e^{0}}{-2}\right) = 1 \Rightarrow k = 2.$$

Therefore the probability density function is

$$f(x) = \begin{cases} 2\,e^{-2x} & \text{for } x > 0 \\ 0 & \text{for } x \le 0 \end{cases}$$

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